MIL-C-24643/58B
3 October 1986
SUPERSEDING
MIL-C-0024643/58A(SH)
8 November 1985
MIL-C-24643/58
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE LS2CS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First	- Copper conductor, tin coated, in accordance with ASTM B 286, size 26-7. Maximum diameter 0.020 inch.
Second	- Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	- Cross-linked polyethylene insulation. Nominal thickness 0.015-inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements. Standard identification code applied by method 6.
Fourth	- Two conductors, one white, one black, cabled together to form a pair in accordance with table I.
Fifth	- The specified number of pairs (see table II) cabled together in accordance with table III. Cabling sequence shall be consecutive, starting with pair no. 1 from center, outward with a left hand lay.
Sixth	- Binder tape applied helically with overlap.
Seventh	- Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.
Eighth	- Spirally wrapped 0.002 inch thick polyethylene terephthalate (mylar) separator tape with 25 percent minimum overlap.
Ninth	- Braided shield of AWG no. 34 tin coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.

(B) denotes changes.

AMSC N/A FSC 6145

DISTRIBUTION STATEMENT A Approved for public release; distribution unlimited

Source: https://assist.dla.mil -- Downloaded: 20261O-20T00:19Z
Check the source to verify that this is the current version before use.

Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket (see table III for thickness). Cable surface marking required.

TABLE I. Pair lay.

Left hand lay

LS2CS-77	LS2CS-60	LS2CS-42	LS2CS-18	LS2CS-6	PAIR NUMBER
				65	62, 68 71,74,77
				66	63,69 72,75
				67	61,64 70,73,76
				53	45,48 51,56,59
				54	43,46,49 52,57,60
				55	44,47 50,58
				27,39	19,22,25 30,33,36,42
				28,40	20,23,26 31,34,37
				29,41	21,24 32,35,38
				13	7,10,16
				14	8,11,17
				15	9,12,18
				1	4
				2	5
				3	6
1.6	3.5	2.5	4.5	2.1	2.9
LENGTH OF LAY (INCHES)					

TABLE II. Details.

Military part no. M24643/58	Type and size	Number of pairs	Cable jacket thickness nominal (inch)	Overall diameter	
				minimum (inch)	maximum (inch)
-01UD	LS2CS-6	6	0.050	0.40	0.43
-02UD	LS2CS-18	18	.050	.55	.59
-03UD	LS2CS-42	42	.050	.75	.80
-04UD	LS2CS-60	60	.065	.88	.93
-05UD	LS2CS-77	77	.085	1.00	1.07

Source: https://assist.dla.mil - Downloaded: 2026-10-20T00:19Z
Check the source to verify that this is the current version before use.

TABLE III. Cable lay.

	LS2CS-6		LS2CS-18		LS2CS-42		LS2CS-60		LS2CS-77	
	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)	No. of pairs	Lay (inches)
Core	6	3.50	6	3.50	1	1.60	6	3.50	3	2.75
Layer 1	-	-	12	6.00	7	4.00	12	6.00	9	5.0
Layer 2	-	-	-	-	14	6.50	18	8.00	15	7.5
Layer 3	-	-	-	-	20	8.50	24	10.00	22	10.0
Layer 4	-	-	-	-	-	-	-	-	28	12.0

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	42.17
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor.....	2000
Conductor to shield.....	1000
Shield to shield, volts, minimum.....	6
Megohms/1000 feet, minimum.....	1.0
Insulation resistance - megohms/1000 feet, minimum.....	500
Jacket flaws.....	No failure
Conductor and shield continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Characteristic impedance - at 1 MHz, ohms.....	135 ± 10
Group B:	
Cold bending cable - at minus 30 ± 2°C, maximum mandrel diameter 12 times cable diameter.....	No damage
Bending endurance - at 25 ± 2°C, cycles, minimum.....	25
Drip - 95 ± 1°C.....	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)..	35
Physicals (unaged)	
Insulation	
Tensile strength - lbf/in^2, minimum.....	1800
Elongation - percent, minimum.....	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum.....	1300
Elongation - percent, minimum.....	160

Source: https://assist.dla.mil -- Downloaded: 202610-20T00:19Z
Check the source to verify that this is the current version before use.

EXAMINATION AND TESTS: (Continued)

	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Permanence of printing (jacket) - cycles, minimum	250
Heat distortion - percent of unaged, maximum	30
Cable filler removability	No failure
Shield-conformance to material, construction and coverage	No failure
Surface transfer impedance	
Milliohms per meter, maximum	700
EMP response time - ms/500 feet	60

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	—
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:19Z
Check the source to verify that this is the current version before use.

UNIT ORDERING LENGTHS:
All sizes 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-58)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:19Z
Check the source to verify that this is the current version before use.